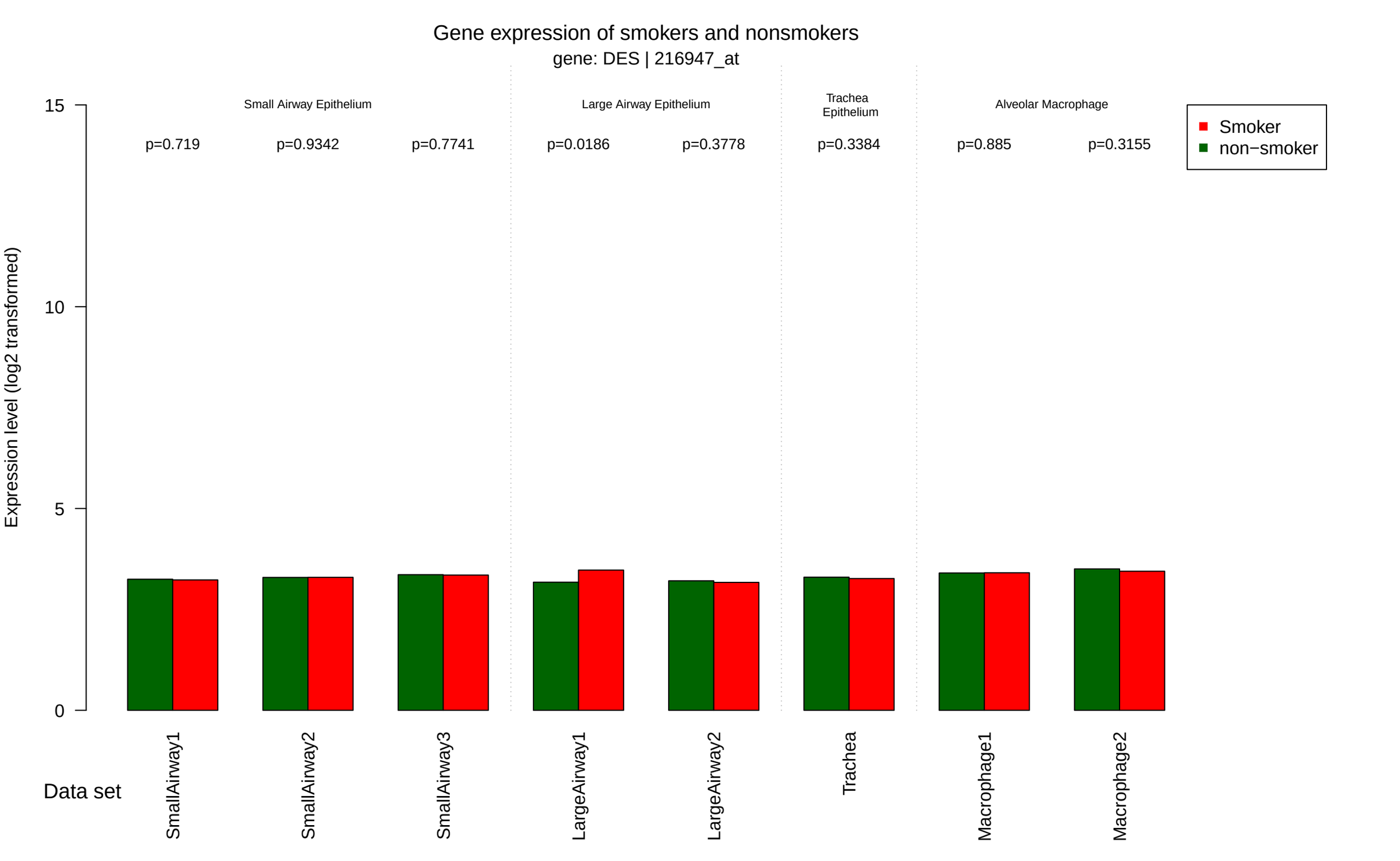
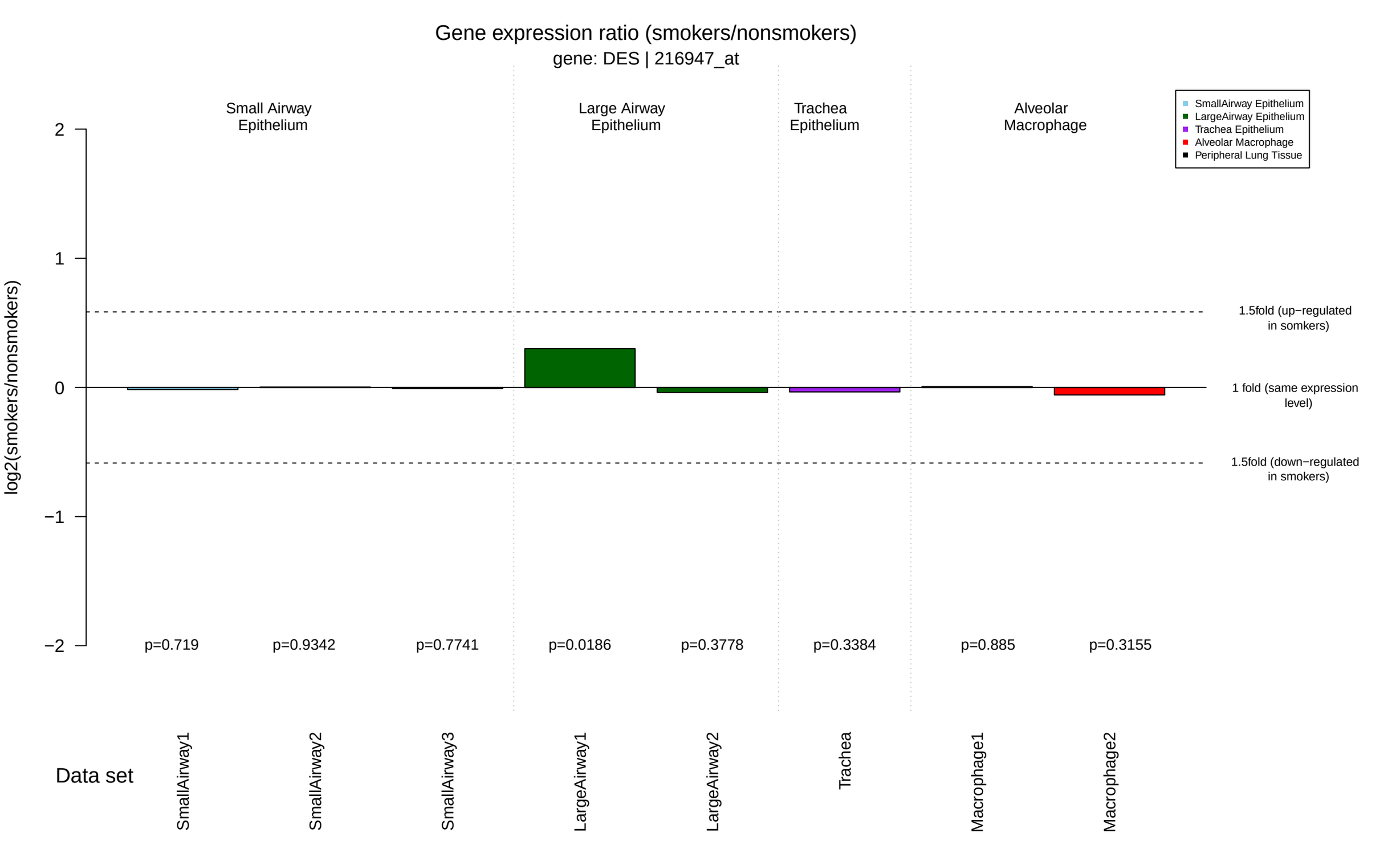
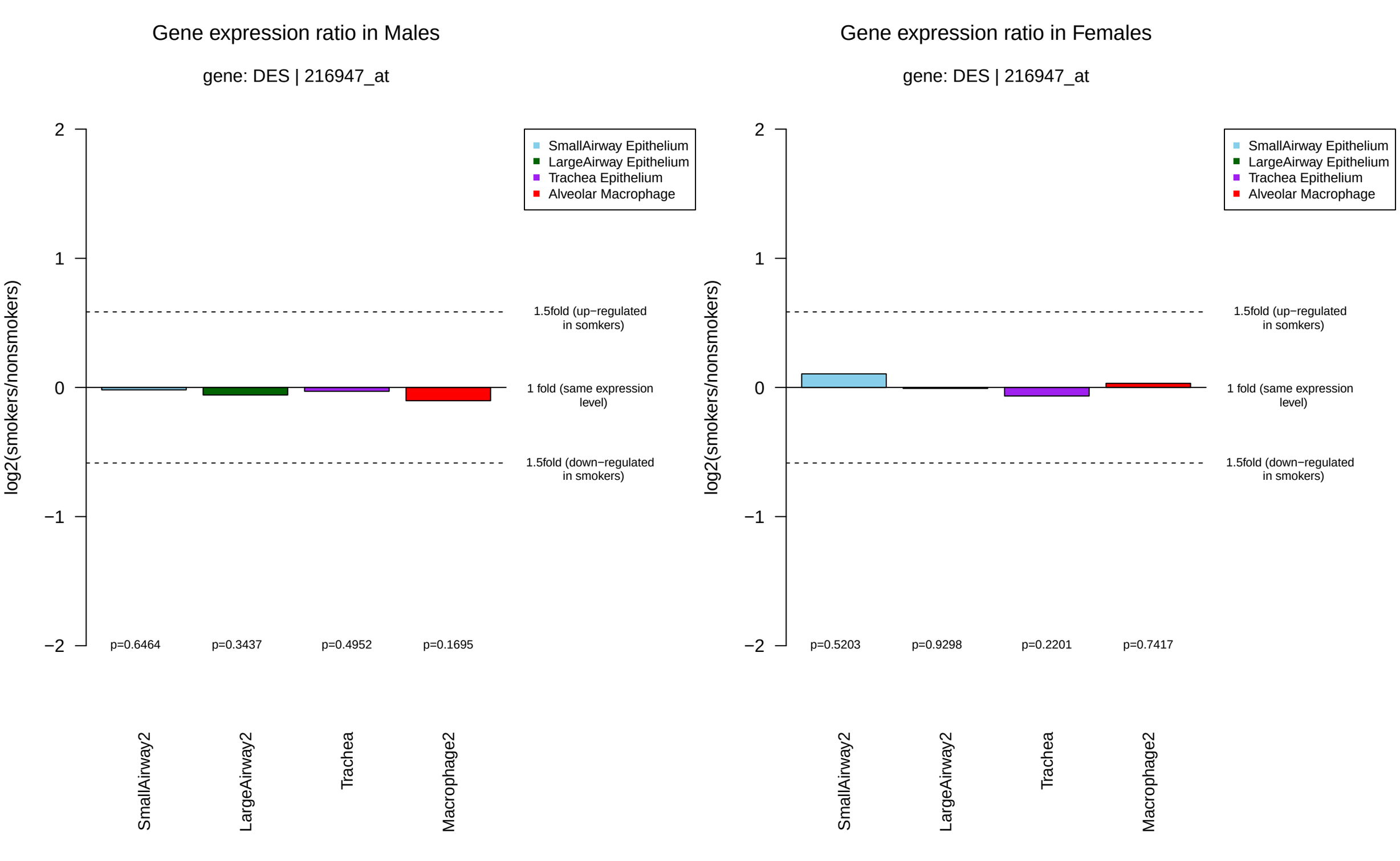
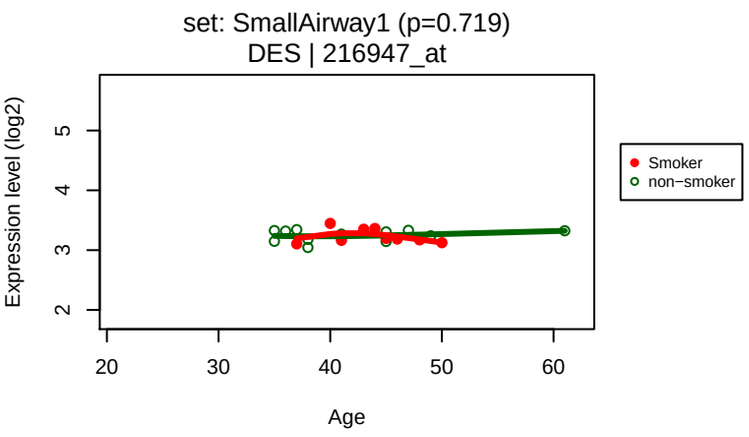
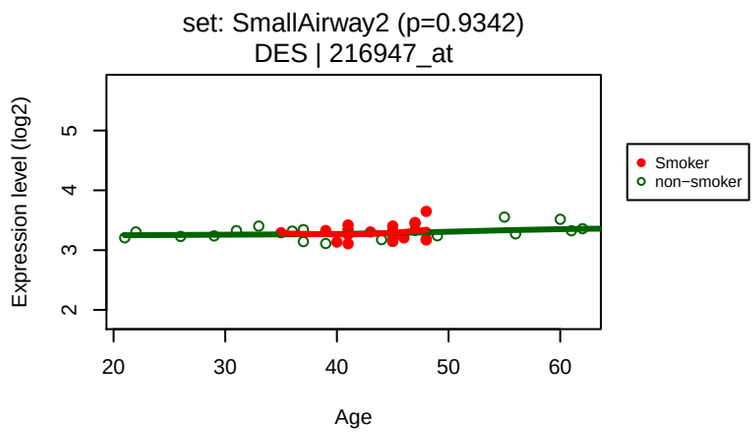
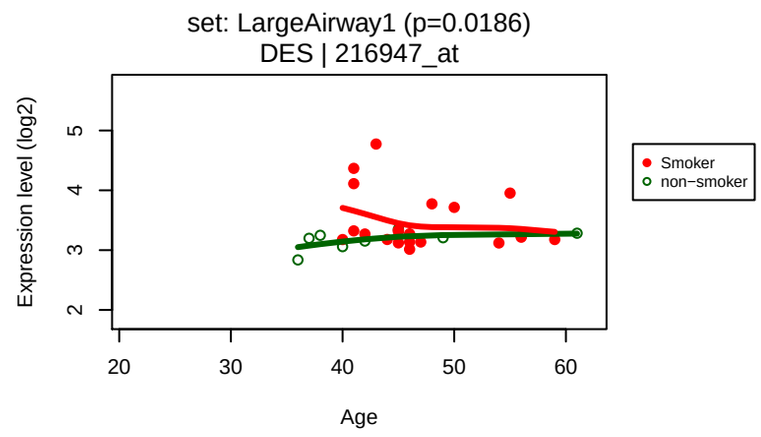

Gene expression of smokers and nonsmokers
gene: DES | 216947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.719
p=0.9342
p=0.7741
p=0.0186
p=0.3778
p=0.3384
p=0.885
p=0.3155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

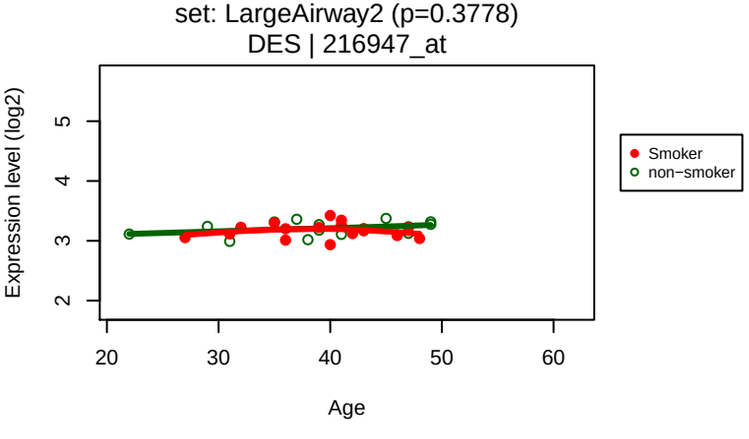

Gene expression ratio (smokers/nonsmokers)
gene: DES | 216947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.719
p=0.9342
p=0.7741
p=0.0186
p=0.3778
p=0.3384
p=0.885
p=0.3155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

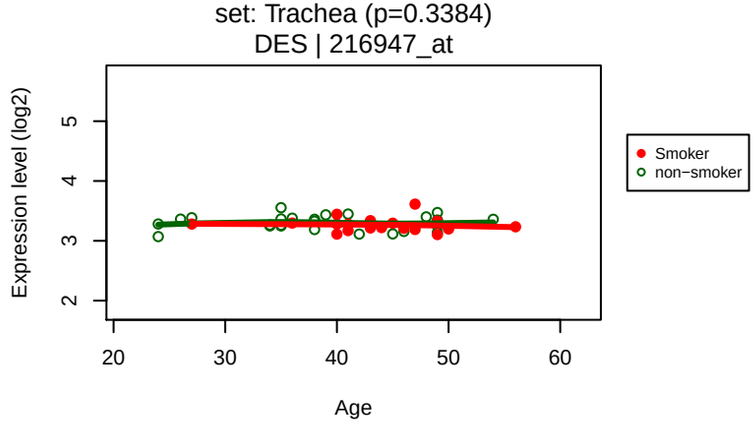

Gene expression ratio in Males
gene: DES | 216947_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6464
p=0.3437
p=0.4952
p=0.1695
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DES | 216947_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5203
p=0.9298
p=0.2201
p=0.7417
SmallAirway2
LargeAirway2
Trachea
Macrophage2

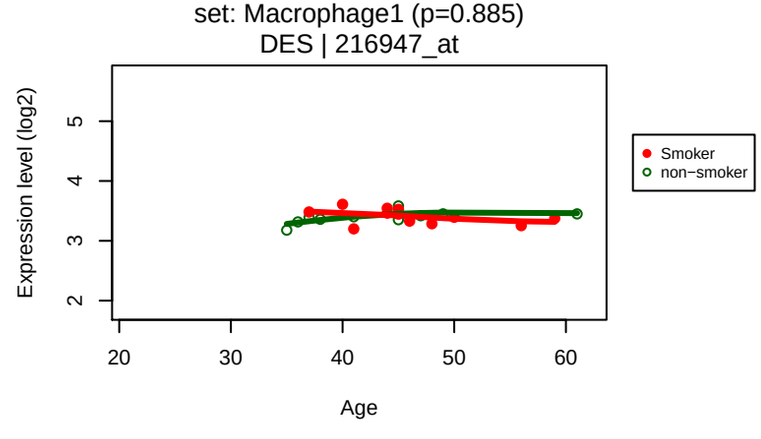

set: SmallAirway1 (p=0.719)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker

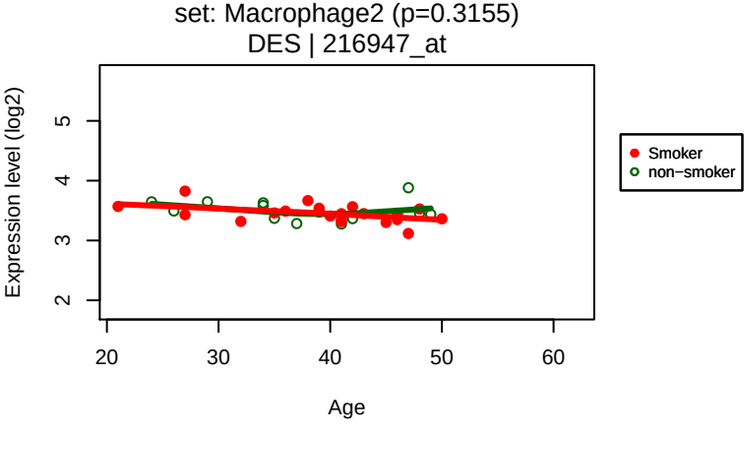

set: SmallAirway2 (p=0.9342)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker

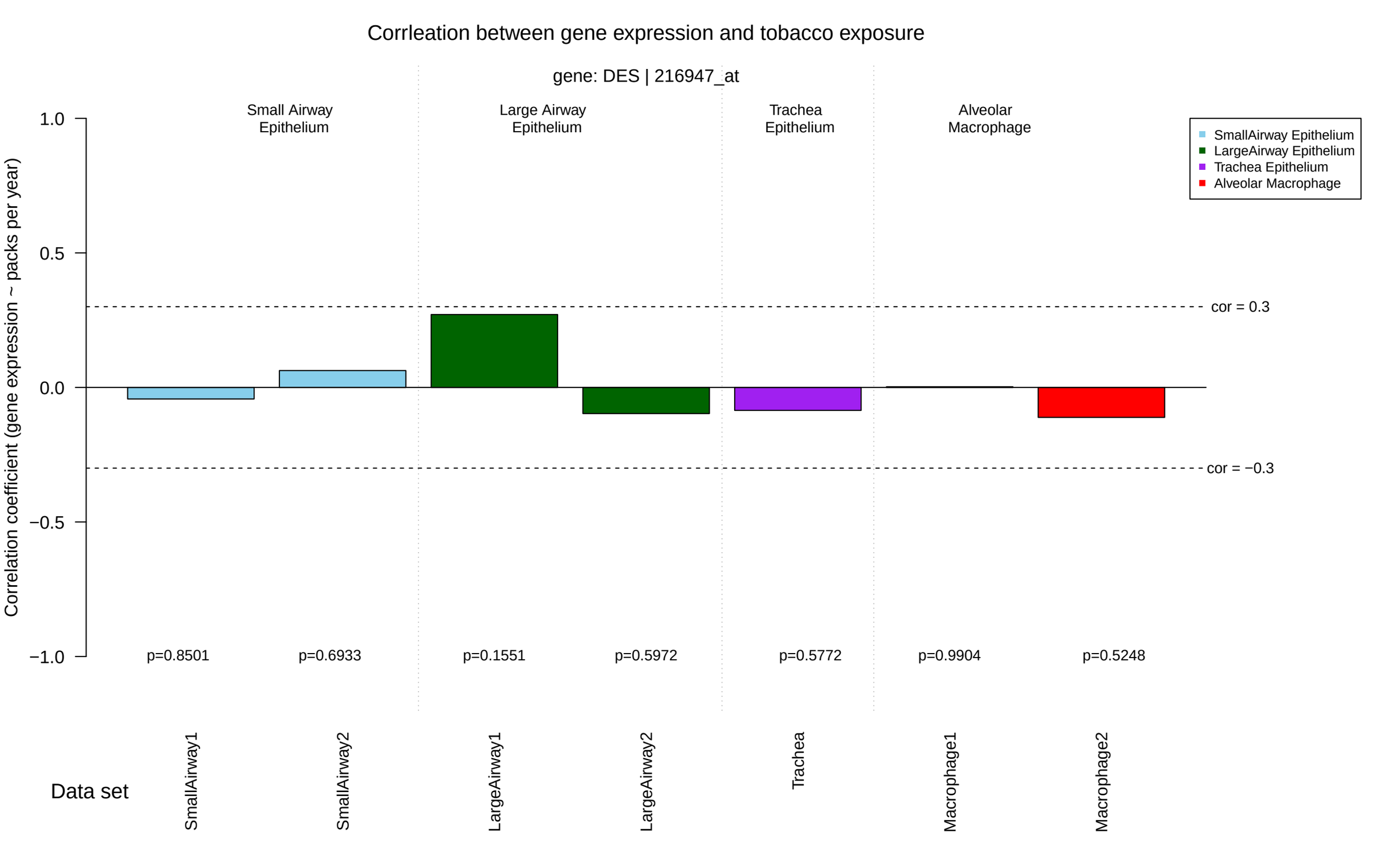

set: LargeAirway1 (p=0.0186)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3778)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3384)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.885)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3155)
DES | 216947_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DES | 216947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8501
p=0.6933
p=0.1551
p=0.5972
p=0.5772
p=0.9904
p=0.5248
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2